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SAYLANI HACKATHON · BEGINNER TRACK

HCM Lite

A small HR web app with HTML, CSS & JavaScript

for Admin and Employees · no framework, no backend



Admin **Employee** HTML · CSS · JavaScript

24 October 2026

Zaitoon Ashraf IT Park

Submit on lms.saylaniit.com

HCM Lite — Beginner Track

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1 The challenge

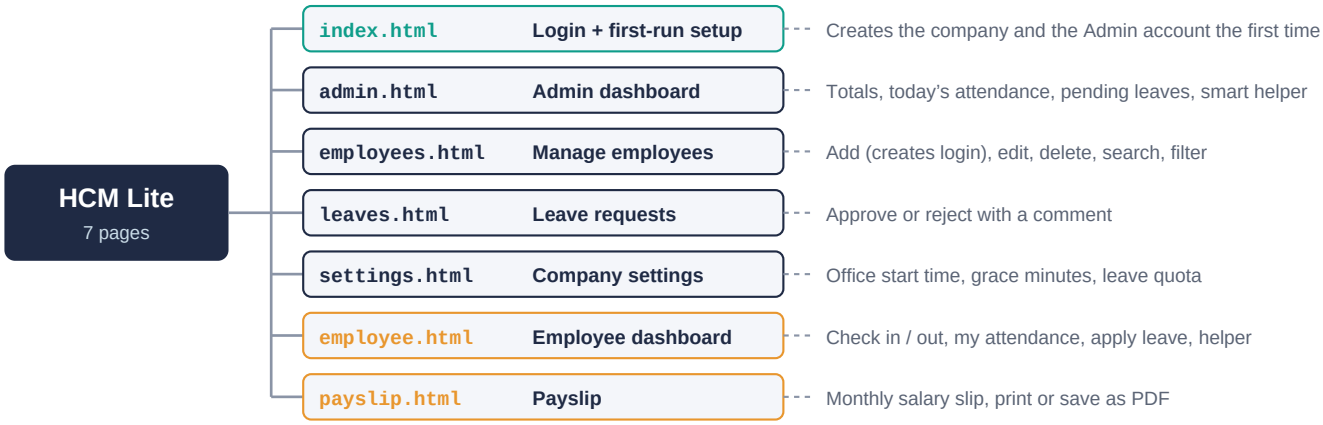
Build **HCM Lite**, a small HR web app, using only **HTML, CSS and JavaScript**. There is no framework and no backend: all data is saved in the browser with **localStorage**. The app starts empty; the Admin sets up the company and adds employees, and every number comes from what people do in the app.

For	Date & venue	Tools	Data	Submit
Students who know basic HTML, CSS and JavaScript	24 Oct 2026, Zaitoon Ashraf IT Park	HTML, CSS, JavaScript, localStorage (Chart.js optional)	Starts empty; no dummy data	Repo + hosted link on lms.saylaniit.com

You must build

- **Two roles**: Admin (runs the company and HR work) and Employee (self-service).
- **Employees**: Admin adds an employee, which also creates that employee's login.
- **Attendance**: Employee checks in and out; the app marks present, late or half day using the Admin's rules.
- **Leave**: Employee applies; Admin approves or rejects; balance updates.
- **Dashboard and payslip**: simple numbers and a monthly salary slip.
- **Smart helper**: a small chat box that answers questions from the app's data (Section 5).
- A **responsive** design that works on a phone, and a live hosted link.

Figure 1 — Pages of the app



Suggested folders: `index.html · pages/*.html · css/style.css · js/storage.js, auth.js, employees.js, attendance.js, leave.js, helper.js`

2 Who can do what

Each page first checks who is logged in. If an Employee opens an Admin page, send them back to their own dashboard.

#	Functionality	Admin	Employee
01	First-run setup: company name and the Admin account	Manage	—
02	Login, logout, change password	Own	Own
03	Company settings: office start time, grace minutes, yearly leave quota, working days	Manage	—
04	Add employee (creates login), edit, delete	Manage	—
05	Employee list with search and department filter	All	—
06	Check in and check out	—	Own
07	Attendance records	All	Own
08	Apply for leave; see status and balance	—	Own
09	Approve or reject leave with a comment	Manage	—
10	Dashboard	Company totals	Personal
11	Monthly payslip (print / save as PDF)	All	Own
12	Smart helper chat	All data	Own data
13	Announcements	Post	Read

Manage = create and change · **All** = every record · **Own** = only their own records · **—** = no access.

3 Business rules (Admin decides them)

Area	Rule
Attendance	Late if check-in is after **office start + grace minutes** (e.g. 09:00 + 15). One check-in per day; check-out only after check-in. Work hours = out – in; under 5 hours = half day. No record on a working day = absent.
Leave	From date must be on or before To date. Count working days only (skip weekends). Cannot overlap another leave. Cannot go over the yearly quota set by Admin.
Payslip	Per day = monthly salary ÷ working days in the month. Deduction = per day × (absent days + 0.5 × half days). Net = salary – deduction.
Forms	Required fields, valid email, email not already used, salary greater than 0. Show the error next to the field.

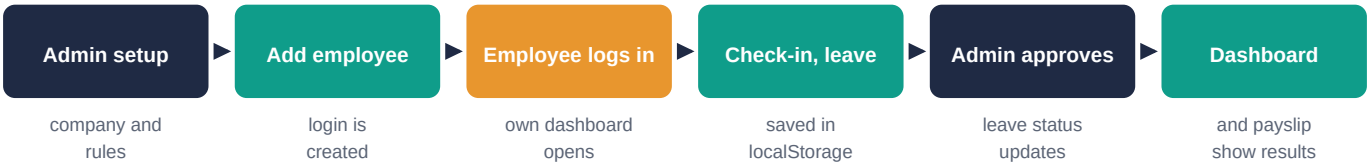
4 Saving data with localStorage

Key	What it stores
hcm_company	Company name and settings (start time, grace minutes, leave quota)
hcm_users	Admin and employees: id, name, email, password, role, department, salary
hcm_attendance	One entry per employee per day: date, check-in, check-out, status
hcm_leaves	Leave requests: employee id, type, from, to, days, reason, status, comment
hcm_session	Who is logged in right now

```
const users = JSON.parse(localStorage.getItem("hcm_users")) || [];
users.push({ id: Date.now(), name, email, password, role: "employee", department, salary });
localStorage.setItem("hcm_users", JSON.stringify(users));
```

Keep these read/write helpers in `js/storage.js` and reuse them on every page. localStorage lives in one browser only, which is fine for learning; real apps use a server and a database.

Figure 2 — How data moves through the app



5 Smart helper (AI touch)

Add a small chat box on both dashboards. The user types a question and the helper answers **from the app's own data**.

Question	Who	Answer comes from
How many employees do we have?	Admin	Count of hcm_users with role employee
Who is on leave today? / Who was late today?	Admin	Approved leaves covering today / today's late records
How many leaves do I have left?	Employee	Quota minus own approved leave days
Was I late today? / How many hours did I work this week?	Employee	Own attendance records

- **Required:** match keywords in the question with JavaScript (e.g. `text.includes("leave")`) and reply with real numbers. Employees only get their own data.
- **Bonus:** connect the helper to a free LLM API with `fetch()`: send the question plus a short summary of the data. Use a demo key only and never commit it to GitHub.

6 Bonus ideas (+10 points)

- Camera check-in: open the camera with `getUserMedia()`, check the location with `navigator.geolocation`, and detect a face with **face-api.js** (CDN).
- Charts with **Chart.js** (CDN), dark mode, export attendance to CSV, a printable monthly report.

7 Scoring and submission

Criteria	Pts	What judges look at
Features work	35	Setup, add employee, login per role, check-in, leave approval, payslip
Business rules	20	Late, half day, leave limits and payslip follow the Admin's settings
Smart helper	15	Correct answers from real data; Employees only see their own
UI and responsive	15	Clean layout, clear forms and errors, works on a phone
Clean code	10	Organized files, reusable functions, readable names
Demo and README	5	Clear demo; README explains how to use the app

Submit on the Saylani LMS: https://lms.saylaniit.com/ — every student shares the **GitHub repo link** and the **hosted link** (GitHub Pages, Netlify or Vercel, all free).

Best of Luck!

Start with login and employees, then attendance and leave. Small steps, working code.

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